# Empirical Project Data Description by Group 18

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**Title:**

## "Which Jobs are Most at Risk in the Age of AI?"

**Research Question:**
How does occupational exposure to artificial intelligence relate to job displacement risk, and does this relationship differ between blue-collar and white-collar workers?

**Data & Samples**
The primary sample is constructed from three cross-sectional waves of the General Social Survey (GSS), conducted in 2021, 2022, and 2024 by NORC at the University of Chicago. The GSS is a nationally representative survey of English- and Spanish-speaking adults living in the United States, covering a broad range of social, economic, and attitudinal topics. The three waves are pooled into a single dataset and restricted to respondents who reported being employed either full-time or part-time at the time of the survey and who had a valid Census 2010 occupation code. Each observation is then matched to an occupation-level artificial intelligence exposure score using the Language Modeling Occupational Exposure (LMOE) index developed by Felten, Raj, and Seamans (2023). The match is performed by linking occupation codes through a Census OCC2010-to-SOC crosswalk.

Additional occupation-level characteristics are merged from the O*NET 30.1 database, published by the U.S. Department of Labor in December 2025. Observations that do not receive an LMOE match after a two-step procedure – consisting of exact SOC matching followed by prefix-based matching – are dropped from the sample. After these restrictions and merges, the final GSS sample contains 5,702 observations.

The unit of observation is an individual employed adult in the United States. Because the GSS is a repeated cross-section rather than a panel dataset, the same individuals are not tracked across survey waves. Instead, each observation corresponds to a different respondent interviewed in the year associated with their record. As a result, the sample is designed to generalize to the population of employed U.S. adults during the period from 2021 to 2024.

A secondary sample is constructed using the Current Population Survey (CPS) Displaced Workers Supplement (DWS), a biennial supplement administered by the U.S. Bureau of Labor Statistics that asks respondents whether they lost or left a job during the previous three years. This sample is restricted to working-age adults between 20 and 65 who have valid occupation codes and who fall within the DWS universe—meaning they were asked the displacement questions. The same LMOE and O*NET merges are applied as in the GSS sample. For displaced workers, the occupation associated with the lost job is used rather than the respondent's current occupation so that the AI exposure measure reflects the pre-displacement role.

Occupational prestige scores are also merged into the CPS dataset using a crosswalk constructed by collapsing mean GSS prestige scores by occupation code. The final CPS sample contains 121,641 observations, of which 112,804 report having been displaced. Whereas the GSS dataset captures how workers perceive the potential threat of AI to their job security, the CPS displaced worker data capture realized job loss outcomes. Together, these two datasets provide complementary perspectives on the relationship between AI exposure and labor market disruption.

**Variables & Description**

Our analysis draws upon the following 15 variables, grouped by their role in the analysis:

**AI Exposure Measures (occupation-level, continuous)**

- **LMOE Score** – Language Modeling Occupational Exposure index (Felten et al. 2023). Measures how much of an occupation's task content can be performed by large language models. Primary proxy for AI exposure.
- **Degree of Automation** – O*NET Work Context rating of how automated the occupation already is (1–5 scale). Direct measure of existing automation intensity.
- **Routine Cognitive Task Index** – Average O*NET importance rating for processing information, documenting, performing administrative tasks, and identifying objects. Captures the task types most directly substitutable by AI.

Table 1A. AI Exposure Measures

| | N | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| LMOE Score | 5702 | 0.27 | 0.89 | -1.68 | 1.93 |
| Degree of Automation | 3194 | 2.15 | 0.45 | 1.12 | 3.80 |
| Routine Cognitive Task Index | 3194 | 3.53 | 0.45 | 2.07 | 4.40 |

**Outcome Variables**

- **Job Loss Risk** – categorical, ordinal. Source: GSS. Asks respondents how likely they are to lose their job in the next 12 months. Takes four values: 1=very likely, 2=fairly likely, 3=not too likely, 4=not likely. Only asked of a random subsample of GSS respondents due to the survey's split-ballot design, yielding approximately 1,162 non-missing

observations in 2024. Serves as a subjective, forward-looking measure of perceived displacement risk.

- **Job Finding Ease** – Job Finding Ease – Categorical, ordinal. Source: GSS. Asks respondents how easy it would be to find an equally good job if they lost or left their current one. Takes three values: 1=very easy, 2=somewhat easy, 3=not easy. Measures perceived labor market vulnerability conditional on displacement occurring.
- **Job Satisfaction** – Categorical, ordinal. Source: GSS. Asks respondents how satisfied they are with their work on the whole. Takes four values: 1=very satisfied, 2=moderately satisfied, 3=a little dissatisfied, 4=very dissatisfied. Workers in highly AI-exposed occupations may report lower satisfaction even before displacement occurs, making this an early warning signal of AI's labor market impact.
- **AI Worry** – Categorical, ordinal. Source: GSS 2024 only. Asks how worried the respondent is that artificial intelligence will take over many of the jobs currently done by humans. Takes five values: 1=very worried to 5=not at all worried. Available for approximately 835 respondents due to split-ballot design. Validates whether respondents in high-LMOE occupations actually perceive AI as a personal threat.
- **Displaced** – Binary indicator. Source: CPS Displaced Workers Supplement. Equal to 1 if the respondent lost or left a job in the past three years, 0 otherwise. This is the primary realized outcome variable – unlike the GSS perception measures, this captures whether displacement actually occurred.
- **Re-employed** – Binary indicator. Source: CPS, derived from current employment status. Equal to 1 if a displaced worker is currently employed, 0 otherwise. Defined only for displaced respondents. Measures labor market recovery after displacement, capturing whether workers in high-LMOE occupations face greater difficulty returning to employment.
- **Wage Change** – Numerical, continuous. Source: CPS. Calculated as the proportional change in weekly earnings between the lost job and the current job, equal to (current weekly earnings minus weekly earnings at lost job) divided by weekly earnings at lost job. Defined only for displaced workers who are re-employed. Negative values indicate a wage loss after displacement. Captures the earnings cost of displacement, testing whether workers in high-LMOE occupations suffer larger wage losses when they find new work.
- **Displacement Reason** – Categorical. Source: CPS (dwreas). Records the stated reason the job was lost: plant or company closed, position or shift abolished, slack work, or other reasons. Critically, position abolished is the category most consistent with structural AI-driven displacement, as opposed to cyclical or firm-level explanations, allowing us to distinguish AI-related displacement from other forms of job loss.

**Moderator**

- **White Collar** – Binary indicator. Source: derived from GSS occupational prestige score. Equal to 1 if the respondent's occupational prestige score is at or above the sample median, 0 otherwise. Defines the blue-collar and white-collar split that is central to our

research question. In the CPS sample, prestige scores are merged from a crosswalk that collapses mean GSS prestige scores by occupation code.

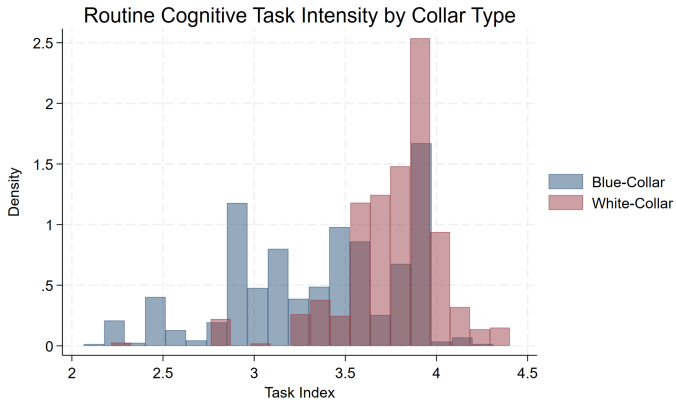
**Occupational Characteristics**

- **Occupational Prestige** – Numerical, continuous. Source: GSS; for CPS, merged from a GSS-derived occupation-level crosswalk. Ranges from 0 to 100, representing the public's assessment of occupational standing based on Census 2010 occupation codes. Used both to construct the white-collar indicator and as a continuous control for occupational rank in regression analyses.

**Individual Controls**

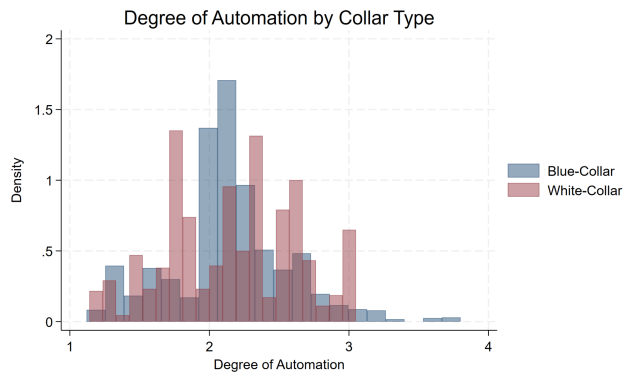
- **Income** – Numerical, continuous. Source: GSS (conrinc). Respondent's own annual income expressed in constant 2024 dollars, adjusted for inflation. Strictly positive among those with valid responses. Higher-income workers may feel more insulated from displacement regardless of their occupation's AI exposure level, making income an important control for separating AI exposure effects from general socioeconomic status.
- **Hours Worked** – Numerical, continuous. Source: GSS (hrs1). Number of hours the respondent worked in the week before the survey, ranging from 1 to 89. Controls for full-time versus part-time employment status, which independently affects both job security perceptions and displacement risk.
- **Age** – Numerical, continuous. Source: GSS and CPS. Respondent's age in years. Older workers may have accumulated occupation-specific human capital that makes them more or less vulnerable to AI displacement, and may also differ in their awareness of and attitudes toward AI technologies.
- **Educational Degree** – Categorical, ordinal. Source: GSS and CPS. Highest educational degree attained, taking five values: 0=less than high school, 1=high school diploma, 2=associate's or junior college degree, 3=bachelor's degree, 4=graduate degree. Education is strongly correlated with both AI exposure – higher-educated workers tend to be in more cognitively demanding occupations – and job security perceptions, making it a critical control variable.
- **Sex** – Categorical, binary. Source: GSS and CPS. Takes two values: 1=male, 2=female. Women are disproportionately represented in clerical and administrative roles that score highly on routine cognitive task intensity and are therefore more directly exposed to language model substitution.
- **Race** – Categorical. Source: GSS and CPS. Takes three values in the GSS: 1=white, 2=Black, 3=other. Racial minorities may have differential access to retraining opportunities and are distributed differently across occupations, making race an important control for interpreting the relationship between AI exposure and job security.

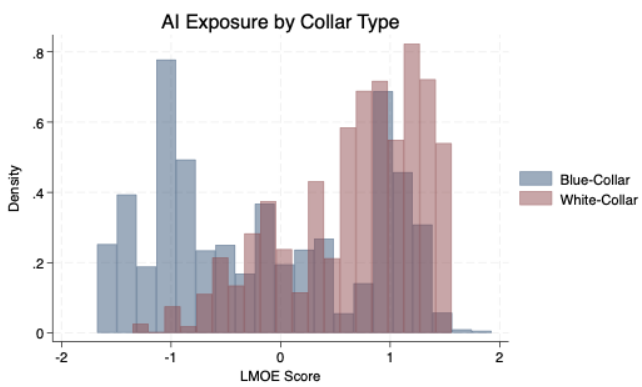
## Informative Figure Analysis



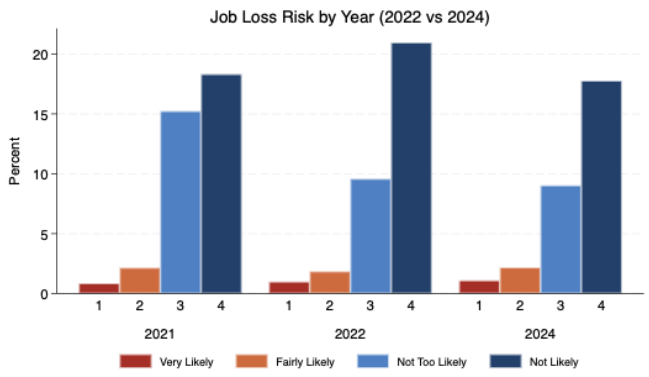

As shown in this graph, there is a visible difference in the cognitive intensity required between white and blue-collar workers. Testing an $H_0$ of $\mu_1 = \mu_2$ returns a p-value of 0.0000, meaning there is also a statistically significant cognitive intensity difference. Given that AI is often better at automating cognitively simpler tasks, this could imply that blue-collar workers are at more risk. However, the intensity difference is also likely to reflect the amount of manual input from workers, which would flip our argument and imply that white-collar workers are at more risk.

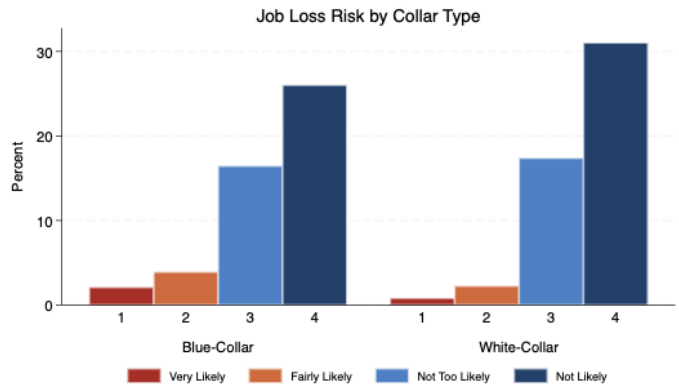
We can look more into the existing automation for blue-collar and white-collar workers in the next graph. There is a less noticeable difference, but t-testing gives us a p-value of 0.0071, showing that white-collar work has a statistically significant increase in automation over blue-collar workers. Using this data with our previous graph, we can see that blue-collar work is indeed more manual, reinforcing our second hypothesis that white-collar workers are at more risk.




We can look at how the Language Model Exposure Score is distributed by blue and white collar workers. We can clearly observe a higher mean LMOE score for white-collar jobs. Indeed, the difference is highly statistically significant, with a p-value $< 0.0001$. This finding is consistent with our hypothesis; white-collar occupations are highly more AI-exposed.




We also compare employees' perception of the labor market, specifically how likely they feel they are to lose their job. Note that the graph on the right includes all the years in the dataset, whereas the graph on left restricts to years 2021, 2022, and 2024 and illustrates the temporal trend. In testing the mean job loss responses between 2022 and 2024, we find that the job loss response mean was higher for 2022 than 2024, and the difference was statistically significant with p-value = 0.027. This means that workers were more worried in 2024 about losing their job than in 2022.

**VARIABLE SHORTLIST (PUT CONTENT ABOVE)**

Sample 1 (GSS):

year, occ10, wrkstat, joblose, jobfind, aiworry, intskill, satjob, prestg10, indus10, conrinc, realrinc, hrs1, hrs2, age, sex, race, educ, degree, marital, childs, region, wrkslf, unemp, compuse, union1, class, finrela, computer_use, ever_unemp, self_emp, union_member, white_collar, lmoe_score, working_w_computers, task_routine_cog, task_nrc_analytic, task_interpersonal, task_manual, degree_automation, repeat_tasks, email_use, time_sitting, time_pressure, indoors_controlled, skill_cognitive, skill_social, skill_tech, knowledge_computers, knowledge_math, knowledge_english, job_zone

Sample 2 (CPS):
year, month, occ2010, dwocc, occ_for_merge, ind, dwind, empstat, labforce, classwkr, dwclass, dwlostjob, dwstat, dwreas, dwrecall, dwnotice, dwyears, dwfulltime, dwweekl, dwwagel, dwunion, dwben, dwexben, dwhi, earnweek, hourwage, uhrsworkt, ahrsworkt, durunemp, whyunemp, jtyears, jtsame, occly, wkswork2, age, sex, race, marst, educ, cinethp, ciemail, cihforwk, cintjbsr, telwrkpay, telwrkhr, wtfinl, displaced, re_employed, wage_change, telework, internet_home, work_computer, white_collar_occ, prestg10, white_collar, lmoe_score, working_w_computers, task_routine_cog, task_nrc_analytic, task_interpersonal, task_manual, degree_automation, repeat_tasks, email_use, time_sitting, time_pressure, indoors_controlled, skill_cognitive, skill_social, skill_tech, knowledge_computers, knowledge_math, knowledge_english, job_zone

Our first sample consists of concatenating cross-sections of the GSS Survey from years 2021, 2022, 2024, which is a nationally representative survey of U.S. adults. The data was then restricted to those working full-time or part-time at the time of the study (*wrkstat* = 1). In addition, we map each respondent's Census 2010 occupation code to a 6-digit SOC code. We also use the Language Modelling Occupational Exposure[1] score, which measures how much a certain job's role can be automated by AI. Variables such as *aiworry* and *intskill* can also proxy for how vulnerable certain jobs are to AI in employees' psychology and measures of skillset. We will run regressions and other statistical models with these variables on other socioeconomic indicators such as unemployment, wage changes, consumer spending, and if unemployed, if the employee expects to be recalled to the job or to find another job.

[1] LMOE (Felten, Raj & Seamans 2023)